A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-77778287})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K, the normalized generator σ_x, the pair (a', J) representing the torsion class e_j, the auxiliary divisor I', the compactly represented element t, a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1-\sigma_x)^2 I' + \operatorname{div}(t) + i_x(J) = 0, \qquad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$-basis of the ideal. For example, the ideal $J = \left(\begin{smallmatrix} 4096 & 3307 \\ 0 & 1 \end{smallmatrix}\right)$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}\,4096 + \mathbf{Z}\,(3307 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Which integral basis. The one PARI returns for the field in question – and that basis is LLL-reduced, so it is not canonical: another machine may reduce differently, and the same coordinate vector would then denote a different algebraic number. The certificate therefore records the basis it was written against, as its last field per entry and in the header for K. The verifier expresses the stored basis in its own, checks that the resulting matrix is integral with determinant ± 1 – so that both bases span the same ring of integers – and converts the coordinates before checking anything. Where the two bases agree, that matrix is the identity and nothing changes.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\mathrm{Cl}(K)$ (as HNF ideals). A character x on $\mathrm{Cl}(K)/5$ is recorded by its value vector $(x(\bar\kappa_1), x(\bar\kappa_2), x(\bar\kappa_3))$; thus $(1, 0, 0)$ is the character dual to $\bar\kappa_1$. The column number j records which basis element e_j of $\mathrm{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x. Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x.

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 19444572 = 0$, of discriminant -77778287. Class group invariants $\begin{pmatrix}120 & 10 & 5\end{pmatrix}$, class number 6000; the three stored generator ideals (HNF) are $\begin{pmatrix}3142 & 2594\\ 0 & 1\end{pmatrix}$, $\begin{pmatrix}48 & 11\\ 0 & 1\end{pmatrix}$, $\begin{pmatrix}56 & 27\\ 0 & 1\end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a, column e_1

Prescribed character vector $\begin{pmatrix}1 & 0 & 0\end{pmatrix}$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 - 1032y^3 + (-35s + 78924)\,y^2 + (1320s - 4800336)\,y + (-15895s + 62297300)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 2060*y^8 + 161941*y^7 - 8849954*y^6 - 19085607*y^5 +
39703209990*y^4 - 2682694921848*y^3 + 88382183865896*y^2 - 141388609
5930480*y + 8792654418192800
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $\begin{pmatrix}0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & -1 & 0\end{pmatrix}$.

The pair (a', J) representing the torsion class

$$a' = \frac{124893499}{288230376151711744} + \left(-\frac{215553}{1152921504606846976}\right) s, \qquad J = \begin{pmatrix}4096 & 3307\\ 0 & 1\end{pmatrix}.$$

$N(J) = 4096$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $356196156263881 = 356196156263881$:

$$\begin{pmatrix}
356196156263881 & 20899795263713 & 240237189096893 & 149682334943811 & 74584464750195 & 8878267596198 & 140590300186640 & 5614782419803 & 26673297946570 & 4201242694612\\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 813 factors with signed exponents of absolute value up to 303826021091825072704778099708288519108721265; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = \begin{pmatrix}0 & 4 & 2\end{pmatrix}$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 2: character a, column e_2

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 - 1032y^3 + (-35s + 78924)\,y^2 + (1320s - 4800336)\,y + (-15895s + 62297300)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 2060*y^8 + 161941*y^7 - 8849954*y^6 - 19085607*y^5 +
39703209990*y^4 - 2682694921848*y^3 + 88382183865896*y^2 - 141388609
5930480*y + 8792654418192800
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad -1 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{5369867}{1803473947459584} + (\frac{37639}{64925062108545024})\,s, \qquad J = \begin{pmatrix} 2304 & 1259 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2304$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1154960917123 = 13^2 \cdot 6834088267$:

$$\begin{pmatrix} 88843147471 & 22981417745 & 74840782850 & 80259555280 & 34361712933 & 48185434100 & 34347972645 & 5446159194 & 55308226386 & 77484776165 \\ 0 & 13 & 7 & 11 & 4 & 0 & 11 & 0 & 6 & 6 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 808 factors with signed exponents of absolute value up to 52731536414621688972007194983297296619005181; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 2 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 3: character a, column e_3

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 - 1032y^3 + (-35s + 78924)\, y^2 + (1320s - 4800336)\, y + (-15895s + 62297300)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 2060*y^8 + 161941*y^7 - 8849954*y^6 - 19085607*y^5 +
39703209990*y^4 - 2682694921848*y^3 + 88382183865896*y^2 - 141388609
5930480*y + 8792654418192800
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad -1 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41}{2809856} + (\frac{5}{550731776})\, s, \qquad J = \begin{pmatrix} 56 & 27 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 56$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $97798082695711 = 163 \cdot 76913 \cdot 7800869$:

$$\begin{pmatrix}
97798082695711 & 43635869222618 & 20248116666304 & 34796842784180 & 74934736526622 & 40600650566810 & 97257713824364 & 25341071896288 & 62800350957332 & 43089669427880 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 813 factors with signed exponents of absolute value up to 2360305956295994705247835682952781364559007 97; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 4: character b, column e_1

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (s + 31)\, y^3 + (18s + 137358)\, y^2 + (-256s + 1550752)\, y + (-1248s - 6966944)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 67*y^8 + 274608*y^7 + 21997344*y^6 + 688521072*y^5 +
15339672784*y^4 + 197835352896*y^3 + 890920697088*y^2 - 918354164633
6*y + 78832002113536
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{124893499}{288230376151711744} + \left(-\frac{215553}{1152921504606846976}\right) s, \qquad J = \begin{pmatrix} 4096 & 3307 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4096$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $5655985846112 = 2^5 \cdot 7 \cdot 2711 \cdot 9313883$:

$$\begin{pmatrix}
706998230764 & 288123381189 & 283434367463 & 266274480700 & 675389283603 & 571519449293 & 2209506510 & 468079798831 & 34771368129 & 448518591696 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 4 & 1 & 0 & 2 & 0 & 0 & 2 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 214 factors with signed exponents of absolute value up to 92778540532570147; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 23$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 5: character b, column e_2

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (s + 31)\,y^3 + (18s + 137358)\,y^2 + (-256s + 1550752)\,y + (-1248s - 6966944)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 67*y^8 + 274608*y^7 + 21997344*y^6 + 688521072*y^5 +
15339672784*y^4 + 197835352896*y^3 + 890920697088*y^2 - 918354164633
6*y + 78832002113536
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{5369867}{1803473947459584} + (\frac{37639}{64925062108545024})\, s, \qquad J = \begin{pmatrix} 2304 & 1259 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2304$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $66221306124176 = 2^4 \cdot 7^3 \cdot 31 \cdot 3301 \cdot 117917$:

$$\begin{pmatrix} 337863806756 & 178223357830 & 303412676812 & 115211076987 & 158684010553 & 148667950759 & 272880645631 & 261550201966 & 96196896100 & 208711196480 \\ 0 & 14 & 7 & 9 & 9 & 4 & 1 & 4 & 12 & 7 \\ 0 & 0 & 7 & 2 & 3 & 2 & 1 & 1 & 5 & 0 \\ 0 & 0 & 0 & 2 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 219 factors with signed exponents of absolute value up to 67327216256039976; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 23$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (1 \quad 0 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 6: character b, column e_3

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (s + 31)\, y^3 + (18s + 137358)\, y^2 + (-256s + 1550752)\, y + (-1248s - 6966944)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 67*y^8 + 274608*y^7 + 21997344*y^6 + 688521072*y^5 +
15339672784*y^4 + 197835352896*y^3 + 890920697088*y^2 - 918354164633
6*y + 78832002113536
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41}{2809856} + (\frac{5}{550731776})\, s, \qquad J = \begin{pmatrix} 56 & 27 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 56$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $5655985846112 = 2^5 \cdot 7 \cdot 2711 \cdot 9313883$:

$$\begin{pmatrix} 706998230764 & 288123381189 & 283434367463 & 266274480700 & 675389283603 & 571519449293 & 2209506510 & 468079798831 & 34771368129 & 448518591696 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 1 & 0 & 2 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 214 factors with signed exponents of absolute value up to 46586216712135294; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 23$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 7: character c, column e_1

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (-s - 4299)\, y^3 + (-64s - 102286)\, y^2 + (-993s - 921900)\, y + (-8290s - 31718952)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8599*y^8 - 204636*y^7 + 36085479*y^6 + 3305291472*y^5 + 13666
2546011*y^4 + 3255416288604*y^3 + 47148827485780*y^2 + 3786568099016
16*y + 2342665576675584
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{124893499}{288230376151711744} + \left(-\frac{215553}{1152921504606846976}\right) s, \qquad J = \begin{pmatrix} 4096 & 3307 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4096$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $9165540440064 = 2^{11} \cdot 3 \cdot 1491787181$:

$$\begin{pmatrix} 71605784688 & 10935834818 & 32893286152 & 49097809214 & 69462646837 & 63458894650 & 16182820619 & 27121101303 & 36721285179 & 64180997900 \\ 0 & 2 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 8 & 4 & 4 & 4 & 6 & 2 & 7 & 3 \\ 0 & 0 & 0 & 2 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 168 factors with signed exponents of absolute value up to 71097681334; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 8: character c, column e_2

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (-s - 4299)\, y^3 + (-64s - 102286)\, y^2 + (-993s - 921900)\, y + (-8290s - 31718952)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8599*y^8 - 204636*y^7 + 36085479*y^6 + 3305291472*y^5 + 13666
2546011*y^4 + 3255416288604*y^3 + 47148827485780*y^2 + 3786568099016
16*y + 2342665576675584
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{5369867}{1803473947459584} + \left(\frac{37639}{64925062108545024}\right) s, \qquad J = \begin{pmatrix} 2304 & 1259 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2304$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $9165540440064 = 2^{11} \cdot 3 \cdot 1491787181$:

$$\begin{pmatrix} 71605784688 & 10935834818 & 32893286152 & 49097809214 & 69462646837 & 63458894650 & 16182820619 & 27121101303 & 36721285179 & 64180997900 \\ 0 & 2 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 8 & 4 & 4 & 4 & 6 & 2 & 7 & 3 \\ 0 & 0 & 0 & 2 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 163 factors with signed exponents of absolute value up to 84897089855; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 9: character c, column e_3

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (-s - 4299)\, y^3 + (-64s - 102286)\, y^2 + (-993s - 921900)\, y + (-8290s - 31718952)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 8599*y^8 - 204636*y^7 + 36085479*y^6 + 3305291472*y^5 + 13666
2546011*y^4 + 3255416288604*y^3 + 47148827485780*y^2 + 3786568099016
16*y + 2342665576675584
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41}{2809856} + \left(\frac{5}{550731776}\right) s, \qquad J = \begin{pmatrix} 56 & 27 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 56$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $68918878802304 = 2^7 \cdot 3^4 \cdot 31 \cdot 613 \cdot 349801$:

$$\begin{pmatrix}
79767220836 & 11438771736 & 50291257638 & 11819489082 & 27409674477 & 59584772934 & 56982509419 & 13055183529 & 49813911300 & 21149380443 \\
0 & 6 & 0 & 0 & 3 & 2 & 5 & 4 & 2 & 5 \\
0 & 0 & 2 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 6 & 3 & 0 & 1 & 3 & 0 & 2 \\
0 & 0 & 0 & 0 & 3 & 2 & 1 & 2 & 0 & 2 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 170 factors with signed exponents of absolute value up to 70557543367; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 2 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 779y^3 + (12s - 6)\, y^2 - 958580y + (-5942s + 2971)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 1558*y^8 - 1310319*y^6 + 4293485972*y^4 - 1854075871724*y^2 +
686536572601367
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{124893499}{288230376151711744} + (-\frac{215553}{1152921504606846976})\, s, \qquad J = \begin{pmatrix} 4096 & 3307 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4096$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $2846349581671 = 31 \cdot 27437 \cdot 3346493$:

$$\begin{pmatrix} 2846349581671 & 1682297369622 & 1678090507568 & 1156047051555 & 1381791799261 & 2228682121179 & 1997309518356 & 2509513399425 & 1533831306167 & 2658402252635 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 583 factors with signed exponents of absolute value up to 403766306269288922389878424379589503; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 2 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $\begin{pmatrix}1 & 1 & 0\end{pmatrix}$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 779y^3 + (12s - 6)\,y^2 - 958580y + (-5942s + 2971)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 1558*y^8 - 1310319*y^6 + 4293485972*y^4 - 1854075871724*y^2 +
686536572601367
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $\begin{pmatrix}0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0\end{pmatrix}$.

The pair (a', J) representing the torsion class

$$a' = \frac{5369867}{1803473947459584} + (\frac{37639}{64925062108545024})\,s, \qquad J = \begin{pmatrix}2304 & 1259\\ 0 & 1\end{pmatrix}.$$

$N(J) = 2304$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $14480298734689 = 14480298734689$:

$$\begin{pmatrix}
14480298734689 & 11835226845201 & 9680565505684 & 10999424029291 & 8360049845517 & 13820489869503 & 2067895638300 & 14435929546168 & 11893746832705 & 8777466945304\\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0\\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 585 factors with signed exponents of absolute value up to 523576813596707540583911148145441942; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = \begin{pmatrix}4 & 1 & 1\end{pmatrix}$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $\begin{pmatrix}1 & 1 & 0\end{pmatrix}$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 779y^3 + (12s - 6)\,y^2 - 958580y + (-5942s + 2971)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 1558*y^8 - 1310319*y^6 + 4293485972*y^4 - 1854075871724*y^2 +
686536572601367
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41}{2809856} + (\frac{5}{550731776})\,s, \qquad J = \begin{pmatrix} 56 & 27 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 56$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $18323541393061 = 139 \cdot 11047 \cdot 11933017$:

$$\begin{pmatrix} 18323541393061 & 16211978754092 & 10664542594448 & 15625899834895 & 14651957082628 & 4839205261943 & 16317380592077 & 17321676063005 & 10063342784477 & 12081030381826 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 588 factors with signed exponents of absolute value up to 48018391632548878275573805794717912 4; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 4 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + 1174y^3 + (-7s - 41214)\,y^2 + (-428s - 2487335)\,y + (16472s + 7531733)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 + 2357*y^8 - 89479*y^7 - 3349517*y^6 - 66773458*y^5 - 3
234338544*y^4 + 339276822953*y^3 + 4644203312214*y^2 - 3116811685932
58*y + 5332684251705713
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{124893499}{288230376151711744} + (-\frac{215553}{1152921504606846976})\, s, \qquad J = \begin{pmatrix} 4096 & 3307 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4096$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $559508918811157 = 7^2 \cdot 101 \cdot 179 \cdot 631591867$:

$$\begin{pmatrix}
79929845544451 & 10599194428984 & 38991785709190 & 26451713065383 & 35221877332430 & 28137851883265 & 67670055032610 & 33852524926092 & 19332471239510 & 4533125194340 \\
0 & 7 & 0 & 0 & 0 & 0 & 5 & 2 & 5 & 4 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 328 factors with signed exponents of absolute value up to 548269805871; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 23$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + 1174y^3 + (-7s - 41214)\, y^2 + (-428s - 2487335)\, y + (16472s + 7531733)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 + 2357*y^8 - 89479*y^7 - 3349517*y^6 - 66773458*y^5 - 3
234338544*y^4 + 339276822953*y^3 + 4644203312214*y^2 - 3116811685932
58*y + 5332684251705713
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{5369867}{1803473947459584} + (\frac{37639}{64925062108545024})\, s, \qquad J = \begin{pmatrix} 2304 & 1259 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2304$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is 300199994586889 = 300199994586889:

$$\begin{pmatrix} 300199994586889 & 8705672519418 & 41485331211154 & 214806510506148 & 174091787454198 & 135619276352018 & 142852669450223 & 102984340738558 & 134641115739755 & 128206395287937 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 325 factors with signed exponents of absolute value up to 663407877150; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 23$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \quad 3 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 3y^4 + 1174y^3 + (-7s - 41214)\, y^2 + (-428s - 2487335)\, y + (16472s + 7531733)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6*y^9 + 2357*y^8 - 89479*y^7 - 3349517*y^6 - 66773458*y^5 - 3
234338544*y^4 + 339276822953*y^3 + 4644203312214*y^2 - 3116811685932
58*y + 5332684251705713
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41}{2809856} + (\frac{5}{550731776})\, s, \qquad J = \begin{pmatrix} 56 & 27 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 56$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $16795112593777 = 7^5 \cdot 999292711$:

$$\begin{pmatrix} 48965342839 & 27598327253 & 509648322 & 10549817481 & 35107981713 & 48486240639 & 43242478697 & 23756125723 & 45072467564 & 37751084465 \\ 0 & 7 & 3 & 0 & 0 & 0 & 6 & 4 & 6 & 2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 7 & 0 & 5 & 2 & 2 & 2 & 4 \\ 0 & 0 & 0 & 0 & 7 & 0 & 4 & 2 & 4 & 4 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 312 factors with signed exponents of absolute value up to 135610748907; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 23$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 1 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 1186y^3 + (32s - 16)\, y^2 + 2895873y + (-1888s + 944)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2372*y^8 + 7198342*y^6 + 13042230716*y^4 + 6036553938433*y^2
+ 69311031564032
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{124893499}{288230376151711744} + \Big(-\frac{215553}{1152921504606846976}\Big)\, s, \qquad J = \begin{pmatrix} 4096 & 3307 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 4096$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $12769279858624 = 2^6 \cdot 31 \cdot 6436128961$:

$$\begin{pmatrix} 798079991164 & 273701518340 & 134745358136 & 362162954989 & 570340753092 & 134158423241 & 187789687113 & 46918813881 & 153744742175 & 176156604510 \\ 0 & 2 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 4 & 2 & 0 & 0 & 0 & 3 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 211 factors with signed exponents of absolute value up to 181941769; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 1 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 1186y^3 + (32s - 16)\,y^2 + 2895873y + (-1888s + 944)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2372*y^8 + 7198342*y^6 + 13042230716*y^4 + 6036553938433*y^2
+ 69311031564032
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{5369867}{1803473947459584} + \left(\frac{37639}{64925062108545024}\right) s, \qquad J = \begin{pmatrix} 2304 & 1259 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 2304$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $75441379741696 = 2^{12} \cdot 18418305601$:

$$\begin{pmatrix} 2357543116928 & 1717651003848 & 2217130224660 & 1608375718980 & 585828959257 & 2120707457686 & 747970803048 & 385329353875 & 1461264682060 & 2287426824477 \\ 0 & 4 & 2 & 0 & 2 & 0 & 2 & 0 & 3 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 236 factors with signed exponents of absolute value up to 2607491709; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 1186y^3 + (32s - 16)\,y^2 + 2895873y + (-1888s + 944)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2372*y^8 + 7198342*y^6 + 13042230716*y^4 + 6036553938433*y^2
+ 69311031564032
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad -1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = \frac{41}{2809856} + (\frac{5}{550731776})\,s, \qquad J = \begin{pmatrix} 56 & 27 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 56$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $92802635251712 = 2^{14} \cdot 18133 \cdot 312371$:

$$\begin{pmatrix}
181255146976 & 161868886458 & 114357199600 & 128944294840 & 10791575898 & 69960206744 & 24557963704 & 28992112172 & 41121986395 & 68259954170 \\
0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 16 & 12 & 2 & 6 & 0 & 3 & 2 & 8 \\
0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 206 factors with signed exponents of absolute value up to 75149591; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 7$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (1 \quad 2 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.